Quantum-Enabled Grid Expansion Planning for Distribution System Energy Networks: A QAOA Approach on the California Test System

Abstract: The energy transition demands rapid expansion and reinforcement of electricity distribution networks to accommodate growing renewable generation and electrified demand. Grid expansion planning is a combinatorial optimization problem involving binary build/no-build decisions for candidate power lines, tightly coupled to network physics through power flow constraints. We present a quantum-enhanced approach to this problem using the Quantum Approximate Optimization Algorithm (QAOA). Starting from the California Test System (CATS), a geographically accurate 8,870-bus synthetic grid, we extract subgrid instances of varying size, formulate the expansion problem as a Binary Linear Program (BLP) minimizing investment cost subject to congestion coverage constraints, and convert it to a Quadratic Unconstrained Binary Optimization (QUBO) problem suitable for gate-based quantum hardware. We demonstrate that QAOA with depth p=1 finds the exact classical optimum on a 6-qubit instance (20-bus subgrid, 4 candidate lines), and identify a 24-qubit instance (30-bus subgrid, 8 congested lines) that sits in the NISQ-feasible regime. Our modular pipeline—implemented in Qiskit and pandapower—provides a systematic framework for benchmarking quantum optimization against classical solvers on realistic power system instances of increasing hardness.

1. Introduction

The global energy transition is driving a fundamental shift in electricity generation from centralized fossil-fuel plants to distributed renewable sources such as solar photovoltaics and wind turbines [1]. This transition substantially alters power flow patterns in distribution networks: flows become bidirectional, variable, and spatially dispersed, placing unprecedented stress on infrastructure originally designed for unidirectional power delivery [4]. Distribution System Operators (DSOs) must therefore plan network expansions—adding new lines, upgrading cables, and installing voltage regulation equipment—to prevent congestion (thermal limit violations) and maintain power quality.

The network expansion planning (NEP) problem is inherently combinatorial: for each candidate reinforcement, the planner makes a binary build/no-build decision, and the quality of each decision depends on all others through the coupled physics of power flow. With n candidate lines, the search space grows as 2^n, making exhaustive enumeration intractable for real-world networks with hundreds or thousands of candidates. Classical approaches rely on mixed-integer programming (MIP) solvers such as Gurobi and CPLEX, which employ branch-and-bound techniques that scale exponentially in the worst case [5]. The network reconfiguration variant of this problem has been proven to be strongly NP-hard [3].

Quantum computing offers a fundamentally different approach to combinatorial optimization. The Quantum Approximate Optimization Algorithm (QAOA) [7] is a variational hybrid quantum-classical algorithm specifically designed for combinatorial problems encoded as Ising Hamiltonians. Recent work has demonstrated QAOA's applicability to power system problems including unit commitment [12], network reconfiguration for loss minimization [11], and energy management in prosumer communities [13]. The Quantum Optimization Benchmark Library (QOBLIB) has established standardized problem classes—including network design, Steiner tree, and topology problems—that are classically hard at sizes amenable to current quantum hardware [5].

In this paper, we develop a complete quantum optimization pipeline for the grid expansion problem, following the benchmarking methodology proposed by Koch et al. [5]: we identify problem instances that are hard for classical solvers, formulate them as QUBOs, and benchmark QAOA against exact classical solutions. Our contributions are:

1. A systematic pipeline from real grid data (CATS [8]) through BLP formulation to QUBO conversion and quantum execution.

2. Identification of problem instances at multiple scales: a 6-qubit instance for algorithm validation and a 24-qubit instance in the NISQ regime.

3. Empirical demonstration that QAOA p=1 achieves approximation ratio 1.0 on the small instance, with analysis of scaling behavior.

2. Problem Definition

2.1. Grid Expansion Planning

Consider a distribution network represented as a graph $G = (\mathcal{B}, \mathcal{E})$ where $\mathcal{B}$ is the set of buses (nodes) and $\mathcal{E}$ is the set of existing lines (edges). Each bus $b \in \mathcal{B}$ has a net power injection p_b (generation minus demand), and each line $e \in \mathcal{E}$ has a thermal capacity $\bar{f}_e$ (maximum power flow in MW).

Under stressed operating conditions (e.g., high renewable generation coinciding with peak demand), some lines become *congested*: the power flow f_e exceeds the thermal limit $\bar{f}_e$. Let $\mathcal{L} \subseteq \mathcal{E}$ denote the set of congested lines, identified through DC power flow analysis.

A set of *candidate lines* $\mathcal{C}$ represents potential new connections that could be built to relieve congestion. Each candidate $e \in \mathcal{C}$ connects two buses, has a construction cost $c_e > 0$, and a rated capacity. The planning problem is to select a minimum-cost subset of candidates that resolves all congestion.

2.2. DC Power Flow Model

We employ the DC power flow approximation, standard for transmission and distribution planning studies [8]. Under DC assumptions (lossless lines, flat voltage profile, small angle differences), the power flow on line $e = (i, j)$ is:

$$f_e = b_e(\theta_i - \theta_j) \tag{1}$$

where $b_e = 1/x_e$ is the line susceptance and θ_i, θ_j are voltage angles at buses i and j. The system of equations $\boldsymbol{B\theta} = \boldsymbol{p}$ relates all bus angles to net injections through the bus susceptance matrix $\boldsymbol{B}$.

2.3. Power Transfer Distribution Factors

Rather than solving the full system of equations for each candidate line combination, we use Power Transfer Distribution Factors (PTDFs) [9]. The PTDF matrix $\boldsymbol{\Phi} \in \mathbb{R}^{|\mathcal{E}|\times|\mathcal{B}|}$ gives the linear sensitivity of each line flow to each bus injection:

$$f_e = \sum_{b\in\mathcal{B}} \Phi_{e,b}\, p_b \tag{2}$$

The PTDF matrix is computed as $\boldsymbol{\Phi} = \boldsymbol{B}_f \boldsymbol{B}_{\text{bus}}^{-1}$, where $\boldsymbol{B}_f$ is the branch-bus susceptance incidence matrix and $\boldsymbol{B}_{\text{bus}}$ is the nodal susceptance matrix (with the slack bus row/column removed for invertibility).

2.4. Influence Score Matrix

Following the approach of Cuenca et al. [4], we compute an *influence score matrix* $\boldsymbol{A} \in \{0,1\}^{|\mathcal{L}|\times|\mathcal{C}|}$ that encodes whether building candidate e significantly reduces congestion on line l:

$$A_{l,e} = \begin{cases} 1 & \text{if } |\Phi_{l,i_e} - \Phi_{l,j_e}| \cdot s_l \cdot \bar{f}_e^{\text{cand}}/S_{\text{base}} > \tau \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

where (i_e, j_e) are the endpoints of candidate e, $s_l = f_l/\bar{f}_l - 1$ is the congestion severity of line l, $\bar{f}_e^{\text{cand}}$ is the candidate's rated capacity, S_{base} is the system base power, and τ is a binarization threshold (set at the 30th percentile of nonzero influence values).

The intuition is that a new line connecting buses with very different PTDF values for a congested line will divert flow away from that congested line, and the effect is stronger when the congested line is more severely overloaded and the candidate has higher capacity.

3. Mathematical Formulation

3.1. Binary Linear Program (BLP)

With the influence matrix computed, the grid expansion problem is formulated as a Binary Linear Program:

$$\min_{\boldsymbol{x}\in\mathbb{B}^{|\mathcal{C}|}} \quad \sum_{e\in\mathcal{C}} c_e\, x_e \tag{4a}$$

$$\text{s.t.} \quad \sum_{e\in\mathcal{C}} A_{l,e}\, x_e \geq 1, \quad \forall l \in \mathcal{L} \tag{4b}$$

$$x_e \in \{0,1\}, \quad \forall e \in \mathcal{C} \tag{4c}$$

The objective (4a) minimizes total investment cost. Constraints (4b) ensure that each congested line is "covered" by at least one built candidate that can relieve its congestion. This is a *weighted set cover* formulation, which is known to be NP-hard [3].

An optional budget constraint $\sum_e c_e x_e \leq B$ can be added to limit total expenditure.

3.2. QUBO Conversion

To solve (4) on quantum hardware, we convert it to a Quadratic Unconstrained Binary Optimization (QUBO) problem. This involves two transformations:

Inequality to equality. Each constraint $\sum_e A_{l,e}x_e \geq 1$ is converted to an equality by introducing slack variables $\boldsymbol{s}^{(l)} \in \mathbb{B}^{k_l}$ in binary expansion:

$$\sum_{e\in\mathcal{C}} A_{l,e}\, x_e - \sum_{j=0}^{k_l-1} 2^j s_j^{(l)} = 1 \tag{5}$$

where $k_l = \lceil \log_2(\sum_e A_{l,e}) \rceil$ is the number of slack bits needed for constraint l.

Penalty encoding. Equality constraints are moved into the objective as quadratic penalty terms:

$$H_{\text{QUBO}} = \sum_{e\in\mathcal{C}} c_e\, x_e + P \sum_{l\in\mathcal{L}} \left(\sum_{e\in\mathcal{C}} A_{l,e}\, x_e - \sum_j 2^j s_j^{(l)} - 1 \right)^2 \tag{6}$$

where $P > 0$ is a penalty coefficient that must be large enough to enforce feasibility but not so large as to swamp the objective. In our experiments we use $P = 10$, determined empirically.

The total number of QUBO variables (qubits) is:

$$n_{\text{QUBO}} = |\mathcal{C}| + \sum_{l\in\mathcal{L}} k_l \tag{7}$$

This conversion is performed automatically by Qiskit Optimization's `QuadraticProgramToQubo` converter [10].

3.3. Ising Hamiltonian

The QUBO is mapped to an Ising Hamiltonian via the substitution $x_i = (1 - Z_i)/2$:

$$H_{\text{Ising}} = \sum_{i<j} J_{ij} Z_i Z_j + \sum_i h_i Z_i + \text{offset} \tag{8}$$

where Z_i are Pauli-Z operators, J_{ij} are coupling coefficients, and h_i are local fields. The ground state of H_{Ising} corresponds to the optimal solution of the original BLP.

4. Quantum Algorithm

4.1. QAOA

The Quantum Approximate Optimization Algorithm (QAOA) [7] is a variational hybrid quantum-classical algorithm that prepares a parameterized quantum state:

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C} |+\rangle^{\otimes n} \tag{9}$$

where $H_C = H_{\text{Ising}}$ is the cost (problem) Hamiltonian, $H_M = \sum_i X_i$ is the mixer Hamiltonian, p is the circuit depth, and $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)$, $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$ are $2p$ variational parameters.

The algorithm proceeds as follows:

1. Initialize all qubits in $|+\rangle^{\otimes n}$ (uniform superposition).
2. Apply p alternating layers of $e^{-i\gamma_k H_C}$ (phase separation) and $e^{-i\beta_k H_M}$ (mixing).
3. Measure in the computational basis to obtain a candidate solution.
4. Use a classical optimizer to update $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ to minimize $\langle\psi|H_C|\psi\rangle$.
5. Repeat until convergence.

We use the COBYLA (Constrained Optimization BY Linear Approximation) optimizer for the classical loop, as it is derivative-free and robust for noisy objective landscapes.

4.2. Algorithm Pipeline

Algorithm 1 summarizes the complete pipeline.

Algorithm 1: Quantum Grid Expansion Pipeline

Input: Grid network $G = (\mathcal{B}, \mathcal{E})$, load scenario, QAOA depth p

Output: Expansion plan $\boldsymbol{x}^* \in \mathbb{B}^{|\mathcal{C}|}$, cost z^*

```
/* Phase 1: Data preparation */
Extract subgrid via BFS from seed bus
Run DC power flow with stressed loads
Identify congested lines L
Generate candidate lines C
Compute PTDF matrix Φ
Compute influence matrix A
/* Phase 2: QUBO formulation */
Build BLP (4) with costs c and influence A
Convert to QUBO (6) with penalty P
Extract Ising Hamiltonian (8)
Solve classically (exact eigensolver) for benchmark z*_class
/* Phase 3: Quantum optimization */
Initialize QAOA with depth p, optimizer COBYLA
Run QAOA to obtain x*_QAOA, z*_QAOA
Compute approximation ratio r = z*_class / z*_QAOA
return x*_QAOA, z*_QAOA, r
```

5. Implementation

5.1. Software Stack

Software Dependencies

- **pandapower** 3.4 [9]: network modeling, MATPOWER import, DC power flow, PTDF computation.
- **Qiskit** 1.x: quantum circuit construction and simulation.
- **Qiskit Optimization** 0.7 [10]: `QuadraticProgram` modeling, BLP-to-QUBO conversion, `MinimumEigenOptimizer` interface.
- **Qiskit Algorithms** 0.4: QAOA and SamplingVQE implementations.
- **Qiskit Aer**: statevector simulator for exact noiseless simulation.
- **NetworkX**: graph algorithms for subgrid extraction and candidate generation.

All quantum simulations use the `StatevectorSampler` primitive for exact expectation values. The pipeline is designed for straightforward migration to IBM Quantum hardware via `qiskit_ibm_runtime`.

5.2. Data: California Test System (CATS)

We use the California Test System (CATS) [8], a geographically accurate synthetic transmission grid comprising 8,870 buses, 10,162 lines, 784 generators, and 2,472 loads. CATS is derived from publicly available geographic data of California's electric infrastructure combined with realistic synthetic parameters. It is provided in MATPOWER format and is suitable for power flow and optimal power flow analysis.

To create problem instances of manageable size for NISQ-era quantum hardware, we extract connected subgrids using breadth-first search (BFS) from a seed bus. The seed bus (bus 18 in our experiments) is selected from a well-connected region with nearby generators and loads, ensuring the subgrid supports converged power flow solutions.

5.3. Subgrid Extraction

Given a seed bus b_0 and target size n, BFS visits buses in order of graph distance until n buses are collected. All lines with both endpoints in the selected set are retained, along with associated loads and generators. An external grid (slack bus) is placed at the first available generator bus to provide the voltage reference.

Table 1: Problem instances extracted from CATS.

Property	Instance A	Instance B
Subgrid buses	20	30
Subgrid lines	24	37
Congested lines $\|\mathcal{L}\|$	1	8
Candidate lines $\|\mathcal{C}\|$	4	6
Load multiplier α	2.0	2.0
Max loading (%)	111.5	349.1
BLP variables	4	6
BLP constraints	1	7
Slack variables	2	18
QUBO qubits n_{QUBO}	**6**	**24**
Ising Pauli terms	16	—

Table 2: Benchmark results on Instance A (6 qubits).

Method	Obj. Val.	Approx. Ratio	Time (s)	Lines Built
Classical (exact)	2.512	1.000	0.01	1
QAOA p=1	2.512	1.000	6.3	1
QAOA p=2	2.512	1.000	15.9	1

5.4. Congestion Simulation

Loads are uniformly scaled by a multiplier $\alpha > 1$ to simulate stressed operating conditions. DC power flow is solved using pandapower [9], and lines with loading exceeding 100% are flagged as congested. In our experiments, $\alpha = 2.0$ produces 1–8 congested lines depending on subgrid size.

5.5. Candidate Line Generation

Candidate new lines are generated between pairs of buses in the congested region (buses adjacent to congested lines plus their 1-hop neighbors) that are not already directly connected. Candidates are filtered to those within 4 graph hops of each other. Costs are differentiated using impedance-weighted shortest path distances as a proxy for construction difficulty.

6. Experimental Results

6.1. Problem Instances

We generate two primary instances from CATS, summarized in Table 1.

Instance A (6 qubits) is small enough for exact statevector simulation and serves as a validation benchmark. Instance B (24 qubits) demonstrates the rapid scaling driven by slack variables: 8 congested lines require 18 slack bits, pushing the problem into the regime where statevector simulation becomes computationally expensive ($2^{24} \approx 16.7$ million amplitudes) and quantum hardware execution becomes attractive.

6.2. Benchmark Results: Instance A

Table 2 presents the results for Instance A.

Table 3: QUBO scaling with grid and congestion complexity.

Buses	$\|\mathcal{C}\|$	$\|\mathcal{L}\|$	Slacks	n_{QUBO}	Class. hard?
20	4	1	2	6	No
25	6	2	6	12	No
30	6	8	18	24	Borderline
30	10	8	18	28+	Yes (MIP gap)

Key Finding: QAOA Achieves Exact Optimum

Both QAOA depths find the exact classical optimum, yielding approximation ratio $r = 1.0$. The optimal plan builds a single candidate line (candidate 0, cost 2.512) connecting buses 10 and 12, which resolves the single congestion on line 5 (111.5% loading between buses 9 and 0).

The classical solver (NumPy exact eigensolver) solves in 0.01 s, while QAOA p=1 requires 6.3 s and p=2 requires 15.9 s on statevector simulation. This overhead is expected for small instances where the classical solver exploits the small Hilbert space dimension ($2^6 = 64$) directly.

6.3. Scaling Analysis

Table 3 shows how the QUBO size scales with problem parameters.

The qubit count grows primarily with the number of congested lines (which drive slack variable count) rather than the number of candidates. This is consistent with the structure of the weighted set cover formulation: each coverage constraint introduces $O(\log |\mathcal{C}|)$ slack bits. For the 30-bus instance with 8 congested lines, the 24-qubit QUBO has a Hilbert space of dimension $\sim 1.7 \times 10^7$, making it a meaningful benchmark for NISQ hardware while remaining simulable on classical computers for validation.

6.4. Discussion

QAOA effectiveness at low depth. QAOA p=1 achieves the exact optimum on Instance A, consistent with theoretical results showing that QAOA can exactly solve certain structured problems at depth 1 when the optimal solution has a simple structure [7]. The single-constraint, well-separated cost structure of Instance A is favorable for QAOA.

Scaling bottleneck: slack variables. The dominant source of qubit growth is inequality-to-equality conversion, not the number of original decision variables. This motivates exploring alternative constraint-handling strategies such as custom mixers [13], Grover-type penalty-free approaches, or decomposition methods that solve subproblems independently.

Toward quantum advantage. Following the QOBLIB methodology [5], practical quantum advantage requires in-

stances where classical solvers exhibit exponential scaling or cannot prove optimality within reasonable time. Our 24-qubit instance (Instance B) is a step toward this regime; scaling to 50–100 candidates with dense congestion patterns would produce instances of 50–200 qubits where MIP solvers are expected to struggle.

7. Conclusion and Future Work

We have presented a complete quantum optimization pipeline for distribution network expansion planning, from real grid data through QUBO formulation to QAOA execution. Our approach uses the CATS synthetic grid to generate problem instances of varying difficulty, formulates expansion as a weighted set cover BLP, and converts to QUBO for quantum hardware compatibility. QAOA achieves the exact classical optimum on a 6-qubit instance derived from a 20-bus CATS subgrid.

Future work will pursue several directions:

1. **Hardware execution:** Run the 24-qubit Instance B on IBM Eagle/Heron quantum processors via Qiskit Runtime, benchmarking against statevector simulation to assess noise effects.

2. **Scaling study:** Systematically identify the crossover point where classical MIP solvers (Gurobi, CPLEX) exhibit exponential scaling on CATS-derived instances, following the QOBLIB protocol [5].

3. **Advanced quantum methods:** Investigate constraint-aware QAOA variants (e.g., FlexQAOA [13] with XY-mixers) to reduce qubit overhead from slack variables.

4. **Multi-objective extension:** Extend to multi-objective optimization balancing cost, reliability, and loss reduction, potentially leveraging quantum multi-objective approaches [6].

5. **MPS simulator comparison:** Demonstrate degradation of matrix product state (MPS) simulation on larger instances as motivation for quantum hardware execution.

6. **Real grid data:** Apply the pipeline to real distribution grid data provided by E.ON at later challenge stages.

References

[1] G. Gust, A. Schlüter, S. Feuerriegel, I. Úbeda, J. T. Lee, and D. Neumann, "Designing electricity distribution networks: The impact of demand coincidence," *European Journal of Operational Research*, vol. 315, no. 1, pp. 271–288, 2024.

[2] G. Duan and Y. Yu, "Power distribution system optimization by an algorithm for capacitated Steiner tree problems with complex-flows and arbitrary cost functions," *Int. J. Electrical Power & Energy Systems*, vol. 25, no. 7, pp. 515–523, 2003.

[3] A. Khodabakhsh, E. Nikolova, and S. Shmoys, "A submodular approach for electricity distribution network reconfiguration," *arXiv preprint arXiv:1711.03517*, 2017.

[4] J. J. Cuenca, M. Vanin, M. U. Hashmi, A. Koirala, H. Ergun, and B. P. Hayes, "Event-informed identification and allocation of distribution network planning candidates with influence scores and binary linear programming," *IEEE Trans. Power Systems*, vol. 40, no. 1, pp. 492–504, 2024.

[5] T. Koch *et al.*, "Quantum Optimization Benchmark Library — The Intractable Decathlon," *arXiv preprint arXiv:2504.03832*, 2025.

[6] A. Kotil *et al.*, "Quantum approximate multi-objective optimization," *Nature Computational Science*, pp. 1–10, 2025.

[7] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," *arXiv preprint arXiv:1411.4028*, 2014.

[8] S. Taylor, A. Rangarajan, N. Rhodes, J. Snodgrass, B. C. Lesieutre, and L. A. Roald, "California Test System (CATS): A geographically accurate test system based on the California grid," *IEEE Trans. Energy Markets, Policy and Regulation*, vol. 2, no. 1, pp. 107–118, 2024.

[9] L. Thurner *et al.*, "pandapower — An open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Trans. Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.

[10] Qiskit Community, "Qiskit Optimization 0.7," 2024. [Online]. Available: https://github.com/qiskit-community/qiskit-optimization

[11] F. F. Silva, P. M. Carvalho, L. A. Ferreira, and Y. Omar, "A QUBO formulation for minimum loss network reconfiguration," *IEEE Trans. Power Systems*, vol. 38, no. 5, pp. 4919–4931, 2023.

[12] S. Koretsky *et al.*, "Adapting quantum approximation optimization algorithm (QAOA) for unit commitment," in *Proc. IEEE Int. Conf. Quantum Computing and Engineering (QCE)*, pp. 181–187, 2021.

[13] Aqarios GmbH, "Quantum energy optimization with FlexQAOA," 2025. [Online]. Available: https://aqarios.com/flexqaoa-energy-optimization-prosumer/